

Principal Component Analysis for Brainwave Activity

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Abstract

We apply principal component analysis (PCA) to the PhysioNet EEG Motor Movement/Imagery Dataset [3], featuring 106 subjects with 64 voltage channels sampled at 160 Hz, in order to test whether different motor tasks produce distinct subspaces when represented as a 2D matrix. Five conditions are recorded as a series of voltage measurements, producing a 2D matrix of shape $X \in \mathbb{R}^{n \times 64}$ where n is the number of epochs per condition (left fist, right fist, both fists, both feet, and rest). We’ve measured pairwise mean principal subspaces angles ranging from 4.6 degrees to 6.6 degrees, substantially below that of a control random subspace (51.24°), indicating that the subspaces produced from measured neural activity are condition invariant. Performing a reconstruction-error analysis shows that a single principal component accounts for 82-85% of all signal energy, with the Eckart-Young theorem [2] guaranteeing that it forms the optimal low rank approximation (per the Frobenius norm). Frequency band PCA in the theta, alpha, and beta waves confirms this condition invariance across all bands, with beta showing the most invariant structures (left vs right fist aligns at 3.6°) while single and pairs of hands reach 21°. This is known in the literature, and consistent with the somatotopic specificity of beta desynchronization [11]. We can conclude that EEG variance is person-specific rather than broadly identifiable through specific motor tasks.

1 Introduction

Electroencephalography (EEG) records brain activity through a series of electrodes placed on one’s head, producing time series of voltage measurements corresponding to brain activity. Understanding these signals with concurrent motor movements is fundamental to understanding how certain physical actions affect brain activity.

One question concerns whether PCA subspaces change between motor tasks. If different motor tasks produce distinct subspaces, then subspace comparison could theoretically be used as a mechanism for brain-computer interfaces, where specific actions could be used to trigger other events across arbitrary human subjects[10]. If they are invariant, then it could be a result of neural activity that varies by person, resulting in great challenges for researchers trying to design tools that leverage these measurements. We investigate this using the PhysioNet EEG Motor Movement Imagery Dataset across three analyses, condition-separated PCA with subspace angle analysis across five task conditions, reconstruction-error analysis (Eckart-Young), and frequency-band PCA across the theta, alpha, and beta channels. **We hypothesize that if EEG signals are dominated by person-specific activity, then different motor tasks will produce highly aligned principal subspaces rather than distinct ones.**

2 Preliminaries

Our analysis is performed on the PhysioNet EEG Motor Movement/Imagery Dataset [3]. The dataset contains recordings from 109 people performing motor activities (define) with 64 scalp electrodes placed along their head. Each subject completes 14 different activities, two baseline tasks (eyes open and closed) and 12 specialized tasks including movements of the

left hand, right hand, both hands, and both feet. Raw recordings were stored in EDF format and loaded using the MNE Python library.

Condition	Epochs
left_fist	4,801
right_fist	4,744
both_fists	4,751
both_feet	4,778
rest	17,803

Table 1: Epoch counts by condition

Each of the epochs is a matrix of voltage measurements, where each of the 64 electrodes sample 160 times per second. A total of 64 channels with 401 different times gives a result of 25664 features per epoch.

3 Methodology

3.1 Condition Separated Principal Component Analysis

The epochs were processed from the data by first loading them with the MNE python library then clipping then from -0.5 to $+2.0$ seconds around each motor task event annotation at 160 Hz, producing fixed-length segments of shape (64×401) per epoch. Three of the subjects (S088, S092, S100) were excluded due to shape mismatches, leaving 106 subjects remaining subjects for the analysis. By time-averaging the epochs, we were first able to collapse them into a 64-dimensional vector. After standardizing the data, we performed a principal component analysis to measure the amount of features for the following percentages of cumulative explained variance: 80% for $K=3$, 90% for $K=11$, 95% for $K=27$ components. Using

the SciPy linear algebra library, we were able to measure the principal angles with `scipy.linalg.subspace_angles`.

Each of the epochs can be represented in the form of a tensor. This tensor has a shape of $(n_{\text{epochs}}, 64, 401)$ and is condensed along the time axis according to the summation $x_{e,c} = \frac{1}{T} \sum_{i=1}^T v_{e,c,t}$ where v is the voltage of channel c at time t , where $T = 401$. This gives a 2D matrix $A \in \mathbb{R}^{n \times 64}$. Before the analysis, we made the decision to normalize each channel using z-score standardization such that larger voltages don't dominate the principal components, and rather, specific information is captured. The approximate equation used was:

$$\tilde{x}_{e,c} = \frac{x_{e,c} - \mu_c}{\sigma_c}.$$

Given the standardized matrix $\tilde{X}$ where $d = 64$, the sample covariance matrix is given by [5]:

$$\hat{\Sigma} = \frac{1}{n-1} \tilde{X}^\top \tilde{X} \in \mathbb{R}^{d \times d}$$

PCA finds the eigenvectors for $\hat{\Sigma}$ above. Equivalently, with the singular value decomposition of $\tilde{X}$, we get:

$$\tilde{X} = \mathbf{U} \Sigma \mathbf{V}^\top.$$

We now have that the columns of $V \in \mathbb{R}^{d \times d}$ are the principal component directions, with the singular values relating to the explained variance. Thus, we have that the total proportion of variance (where $\lambda_i = \frac{\sigma_i^2}{n-1}$) [5]:

$$\pi_i = \frac{\lambda_i}{\sum_{j=1}^d \lambda_j}$$

We can then proceed with the component selection, choosing a number of K components according to the amount of explained variance we expect. Cumulative explained variance is given by $\text{CEVR}(k) = \sum_{i=1}^k \pi_i$, implemented quickly with `np.cumsum(pca_temp.explained_variance_ratio_) * 100`.

To compare the principal angles between the subspaces, we use the formula given by Bjorck and Golub [1]. Given the two orthonormal bases of the subspaces, the number K principal angles are defined via the singular values of the cross-product matrix [1]:

$$\cos(\theta_i) = \sigma_i(\mathbf{U}_A^\top \mathbf{U}_B)$$

($U_a^\top$ doesn't have to be the conjugate transpose because it is real-valued). Thus, the mean angle is: $\bar{\theta} = \frac{1}{K} \sum_{i=1}^K \theta_i$ as a scalar per the condition pair.

The sensitivity analysis (just the sample variance between the measurements) is given as given with the time variance per channel, below [5]:

$$s_{e,c}^2 = \frac{1}{T-1} \sum_{t=1}^T (v_{e,c,t} - \bar{v}_{e,c})^2$$

computed as `.var(axis=2)`. The concatenated mean and variance features was built with `np.concatenate([X_mean, X_var], axis=1)`, producing a 128 dimensional vector that was processed with the above pipeline. This was included in order to test whether the combined features have condition specific information, producing larger subspace angles than either of the two subspaces alone.

3.2 Reconstruction

Relative Frobenius reconstruction error was computed for $k \in \{0, 1, 3, 5, 10, 27\}$. The Eckart-Young theorem states that we can make rank- k "truncated" approximations of the following form [2]:

$$\mathbf{X}_k = \mathbf{U}_k \Sigma_k \mathbf{V}_k^\top$$

such that they limit the reconstruction error across all rank $\leq k$ matrices B (if some singular values are zero then the rank could be less than k).

$$\mathbf{X}_k = \min_{\text{rank}(\mathbf{B}) \leq k} \|\mathbf{X} - \mathbf{B}\|_F$$

The Frobenius norm, which we will use to quantify this value that Eckart-Young says gives "less" distance, can be defined as [9]

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i,j} a_{ij}^2} = \sqrt{\sum_{i=1}^r \sigma_i^2}$$

where σ is apparent from the fact that the diagonal matrix Σ represents the same transformation just in a different coordinate system. Thus, the relative matrix norm should be the same.

This leaves a residual after removing the top- k components of the form:

$$\|\mathbf{X} - \mathbf{X}_k\|_F = \sqrt{\sum_{i=k+1}^r \sigma_i^2}$$

Thus, we have the relative Frobenius error as:

$$\varepsilon(k) = \frac{\|\mathbf{X} - \mathbf{X}_k\|_F}{\|\mathbf{X}\|_F}$$

with the residual being:

$$\mathbf{X}_{\text{residual}} = \mathbf{X} - \mathbf{X} \mathbf{V}_k \mathbf{V}_k^\top.$$

This is more easily accomplished with the python snippet `np.linalg.norm(residual, 'fro') / np.linalg.norm(X, 'fro')`.

3.3 Frequency Band Principal Component Analysis

A fourth-order Butterworth bandpass filter (using `scipy.signal`) isolated different frequency channels within the data, including theta (4-8 Hz), alpha (8-13 Hz), and beta (13-30 Hz) [11, 8], Gamma (30+ Hz) was originally planned but dropped after a Welch power spectrum check revealed a sharp roll-off above 40 Hz from prior hardware low-pass filtering, leaving no usable signal in that range. The variance in the channels (band power) replaced the time-mean as the feature being analyzed. Subsequently, subspace angles were computed per band at $K=27$ and $K=5$.

After filtering, each epoch is collapsed to a 64-dimensional feature vector by computing per-channel variance (`epochs.var(axis=2)`). Band-filtered oscillations average to

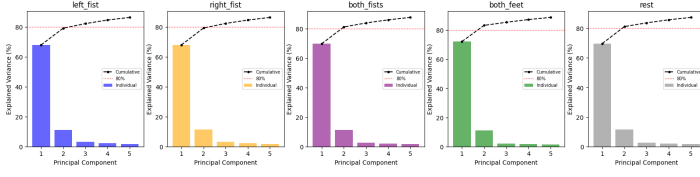


Figure 1: Scree plot for each condition

approximately zero over time, making the time-mean used in the broadband pipeline uninformative for narrow bands. Per-channel variance within a band is the standard EEG measure of band power [6, 8]: it captures how strongly each channel oscillates within that band per epoch, which is the physiologically meaningful quantity. After bandpass-filtering, the zero-centered signal variance gives the band power.

4 Results

4.1 Condition Separated PCA, Component Selection and Subspace Angles.

The first component explains 68-72% of the variance when time-averaging across all five conditions, with 95% reached with $K=27$ features added. All 10 of the pairwise subspace angles fall between 4.6° and 6.6° , indicating substantially similar dominant subspaces regardless of the motor task. Preserving the time structure with a flattening analysis gives a mean angle of 18.5° and 20.0° . See figure 1.

Critically, flattened epochs produced nearly identical results to time-averaging (mean 18.5° vs 20.0°), confirming that the temporal information discarded by time-averaging does not meaningfully affect subspace structure at this level of analysis. The difference between scree plots reveals an important finding: time-averaging forces variance to concentrate in a small number of components, creating an appearance of low dimensionality. The flattened analysis shows the true complexity of EEG, where variance is spread across many spatial-temporal patterns rather than dominated by one or two components. Despite this structural difference, both approaches converge on the same subspace angle results, suggesting the similarity between conditions is a genuine property of the data rather than an artifact of temporal summarization.

The sensitivity analysis shows time variance gives larger angles than the time mean, suggesting that the variance is captured through condition-specific information though still small in absolute terms.

Our total results are quantified in Table 4.1 below.

Condition	PC ₁	
left_fist	67.9	95.0
right_fist	68.0	95.0
both_fists	69.8	95.7
both_feet	72.2	96.1
rest	69.7	95.4

Table 2: PC₁ explained variance and cumulative variance at $K = 27$

Pairwise subspace angles across all 10 of the condition pairs

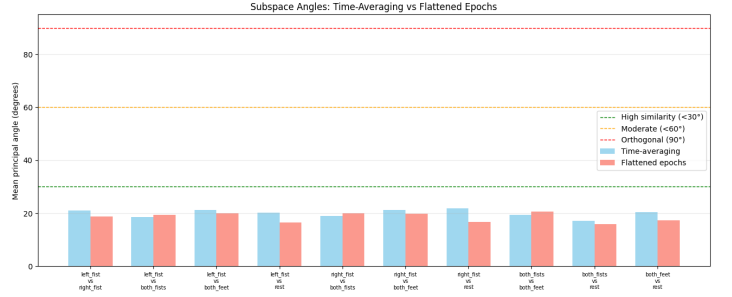


Figure 2: PCA Flattening

are reported in table 3. All angles fall between 4° and 6° , well below orthogonality; this indicates that they are largely condition invariant.

Condition pair	Mean principal angle
left_fist vs right_fist	4.6°
left_fist vs both_fists	6.1°
left_fist vs both_feet	6.3°
left_fist vs rest	5.5°
right_fist vs both_fists	5.9°
right_fist vs both_feet	5.9°
right_fist vs rest	4.7°
both_fists vs both_feet	6.6°
both_fists vs rest	5.3°
both_feet vs rest	5.5°

Table 3: Mean principal angles between condition subspaces ($K = 27$, time-averaged features).

To contextualize these results, we were able to generate 10000 pairs of random K -dimensional subspaces by QR-decomposing Gaussian random matrices [4], computing their mean principal angles. That random baseline yielded 51 degrees. Our observed mean of 5.64 degrees sits substantially below this baseline, confirming that our condition subspaces are not a result of random chance.

The sensitivity analysis is to test the different results achieved with different feature representations. Time-variance gave the largest angles than time mean (20° vs 5° , respectively), suggesting that variance captures more condition-specific information (yet both are below the angle produced by the combination of randomly generated matrices). Additionally, the mean and variance concatenation suggests the mean dominates the combined features.

Feature	Left vs Right	Left vs Rest
Time-mean	4.60°	5.45°
Time-variance	20.15°	22.35°
Mean + Variance	5.14°	7.53°

Table 4: Sensitivity analysis: mean principal angles at $K = 27$ across three feature representations (left vs right and left vs rest only).

Comparing the two approaches, time averaging produced a mean angle of 20 degrees across all pairs, while the flattened IncrementalPCA (preserving the 25,664 dim structure) produced 18.5 degree mean angle. The near identical result

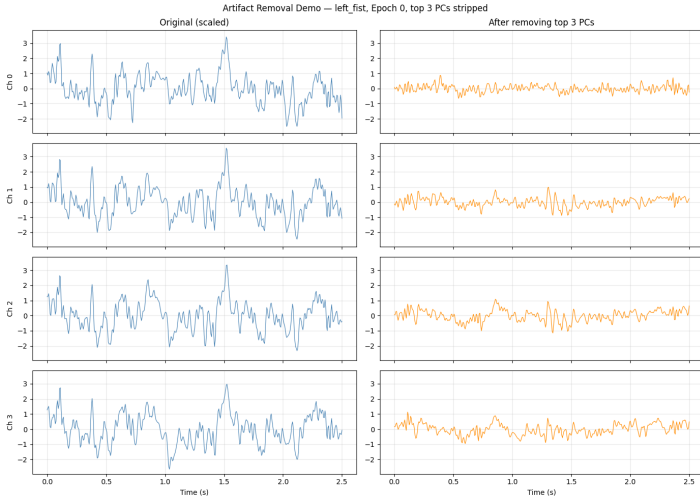


Figure 3: Reduced Variance

across the methods confirm that the conditions are largely invariant and not a result of our chosen feature representation method.

4.2 Reconstruction

Removing the first principal component alone increase error to 0.82-0.85, suggesting that one component accounts for almost all of the signal. Removing the top three principal components raises the error to 0.910.92, and for top 27, it reaches 0.97-0.98. The Eckart-Young discrepancy was 0, confirming that the rank- k approximation is optimal (that is, no other matrix can form a matrix closer to X as measured by the Frobenius norm).

Table 5 and Figure 3 the reconstruction error curve and show that removing the first principal component eliminates 82-85% of the total variance. For $K = 3$, approximately 91% – 92% of the energy is removed, and for $K = 27$, approximately 97% – 98%.

Condition	$k = 1$	$k = 3$	$k = 5$	$k = 10$	$k = 27$
left_fist	0.824	0.906	0.929	0.947	0.975
right_fist	0.824	0.908	0.930	0.947	0.975
both_fists	0.835	0.916	0.936	0.953	0.978
both_feet	0.850	0.924	0.942	0.957	0.980
rest	0.835	0.915	0.935	0.950	0.977

Table 5: Relative Frobenius reconstruction error $\varepsilon(k)$ after removing top k principal components per condition.

4.3 Frequency-Band PCA

Figure 4 show that the mean subspace angles per band for $K = 27$ and $K = 5$. At $K = 27$, all of the angles remain below the random-subspace measurements across the bands and pairs of conditions.

At $K = 5$, the same comparisons span 3.6° – 30.4° with clearer groupings between condition types. This wider dynamic range arises because band-filtering plus per-channel

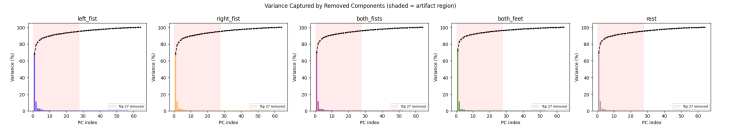


Figure 4: Variance from Removed Components

variance concentrates signal into very few components; using K beyond the effective rank measures noise alignment rather than signal structure. We therefore treat $K = 5$ as the primary analysis for frequency bands, and report $K = 27$ for direct comparability with the broadband results.

At $K = 27$, all 30 condition-pair angles fall in a narrow 10 – 24 range; the noise tail (PCs 6–27, 5 – 15% of variance) dilutes the dominant components. At $K = 5$ the same comparisons span 3.6° – 30.4° with clear groupings. When a feature space has low effective rank, using K beyond the effective rank measures noise alignment, not signal.

At $K = 5$, beta splits into two distinct groups: highly aligned pairs (3.6° – 9.0°) including left_fist vs right_fist (3.6°), single-hand vs rest ($\sim 9^\circ$), and both_fists vs both_feet (7.8°); and less aligned pairs (19.9° – 21.2°) comparing single-hand to bilateral movements. This is consistent with Pfurtscheller and Lopes da Silva (1999) [11], who showed that beta event-related desynchronization is somatotopically specific and localized near sensorimotor areas: conditions engaging similar cortical territories (left vs right fist) produce highly aligned subspaces, while conditions engaging different territories (single-hand vs bilateral) produce larger angles. At $K = 27$ this structure is diluted by noise-dominated components and disappears. Voluntary movement produces a circumscribed event-related desynchronization (ERD) in the lower beta band, localized close to sensorimotor areas; ECoG recordings show beta ERD topography is more discrete and somatotopically specific than alpha ERD [11]. Beta ERS shows somatotopic organization, with distinct cortical loci for finger, arm, and foot movements.

Alpha (8–13 Hz) shows weaker but interpretable structure: single-hand-vs-rest angles ($\sim 15^\circ$) exceed beta’s, and bilateral-vs-rest angles are larger still (15.6° – 19.2°). This fits alpha (mu rhythm) being a more diffuse, attentional rhythm whose desynchronization is broader and less spatially precise than beta’s [11].

Theta (4–8 Hz) acts as a negative control: angles span 12.3° – 30.4° without the clean grouping seen in beta. Theta is not a motor rhythm, so the unstructured pattern confirms the pipeline finds genuine band-specific structure rather than method artifacts.

5 Applications

We believe that the results of this analysis have a number of practical implications for how PCA techniques can be used in EEG research and its related applications. Since we found that the dominant principal subspaces remained highly aligned across all of the motor conditions, our findings would suggest that most of the large-scale variance that was captured by PCA reflects subject-specific brain activity rather than the differences between individual movement tasks. This

finding has important consequences for how these PCA-based features are going to be interpreted in both neuroscience research and further engineering applications.

One of these important applications relates to brain-computer interface (BCI) systems. Many BCI approaches make attempts to distinguish between imagined movements (such as left-hand versus right-hand motion while using EEG recordings). If different motor tasks happened to produce clearly separated PCA subspaces, then it could be possible that the subspace comparisons themselves could potentially be used as a classification tool. However, due to the fact that the observed principal angles between the condition subspaces were consistently small (at about 4° – 6°), our results would perhaps indicate that broadband PCA alone is not going to prove sufficient enough to reliably distinguish between these tasks. Instead, we believe that this supports the idea that effective BCI systems should rely on more of the localized or frequency-specific features, such as more traditional sensorimotor rhythm changes characteristic of the alpha and beta bands.

Our group believes that a second plausible application would concern the signal compression and ensuing artifact removal. For example, the reconstruction-error analysis showed that a single principal component could account for roughly 82%–85% of the signal energy across conditions, and that relatively few components also manage to capture nearly all of the remaining variance. When looking at the Eckart–Young theorem, these components form the optimal low-rank approximation of the data in the Frobenius norm. This would make PCA a useful preprocessing step for reducing the dimensionality, removing the noise, and then simplifying the EEG signals before any form of further analysis.

We find that a third application comes from the frequency-band PCA results. In particular, we found that the beta band was showing a clearer condition-related structure than was found in the theta band, which is consistent with the existing neuroscience literature that describes the role of beta desynchronization in motor activity. To us, this would suggest that these subspace-angle comparisons within specific frequency ranges may be more informative than broadband analyses when it comes to studying movement-related brain activity. As a result, it is possible that these sorts of band-limited PCA methods may prove to be especially useful in clinical monitoring and neurofeedback settings where tracking oscillatory activity within any sort of particular frequency ranges is important.

Overall, we believe that these results show that PCA-based subspace methods are most useful when it comes to understanding the subject-level structure, reducing the dimensionality of the EEG feature space, and identifying frequency-dependent neural patterns, rather than just directly separating the motor-task conditions using the broadband and the EEG data alone.

6 Conclusion

Across all three analyses, the dominant EEG subspace is invariant across all conditions. Different motor tasks produce pairwise principal angles below 7 degrees in the broadband

analysis, and below 25 degrees across all frequency bands (not near orthogonal). This is consistent with Kumar (2019) [7] who showed that person-specific neural activity dominates subspace structure regardless of task [7]. We established a random sequence baseline, ultimately confirming that the observed mean angle is below chance and reflects genuine subspace invariance rather than deriving from other sources. Furthermore, one principal component accounts for roughly 83% of signal energy across all conditions, with the Eckart–Young theorem guaranteeing that removing it is the optimal Frobenius-norm compression. Further research should investigate single-trial ERD/ERS features which may capture motor task specific information obscured by the dominant shared subspace.

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Appendix

Here is a link to our code: [Google Drive](#).

Notation

The Eckart-Young Paper uses different notation. They denote the matrix to be approximated as α and its lower-rank approximation as β , whereas we use $\mathbf{X}$ and $\mathbf{X}_k$. Their scalar product of two matrices, defined as $(\alpha, \beta) = \sum_i \sum_j \alpha_{ij} \beta_{ij}$, is equivalent to what is described above as the “Frobenius norm.” Their conclusion $\alpha = u' \lambda U$, corresponds to our singular value decomposition $\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$, with their orthogonal matrices u, U mapping to our $\mathbf{U}, \mathbf{V}$ and their diagonal matrix λ mapping to our $\mathbf{\Sigma}$. Finally, their rank- r approximation is obtained by retaining only the r largest diagonal elements of λ and zeroing the other diagonal elements which is exactly the truncated SVD we describe as $\mathbf{X}_k = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top$.